Emerging Energy Harvesting Techniques For Low-Power No-Battery Solutions

Ultra-low-power techniques are emerging for operation of devices without dependence on batteries. Will such emerging energy harvesting techniques be the key for sustaining low-power devices? Will the devices be feasible for their applications? If so, how?

Darshil Patel

Most portable electronic devices we use are battery powered. However, innovative solutions for battery-less designs are being constantly developed by researchers and product designers. It is the beginning of a new era, where the latest wireless electronics show the best of both the worlds by being portable and battery-free.

Researchers across the globe have figured out various ways to harvest electricity from ambient environments and power devices. Recent studies have suggested that a significant amount of power can be harnessed from surroundings to power low-power electronic devices like IoT sensors, actuators, etc.

The technologies discussed here enable battery-less solutions and how researchers and design engineers are progressing towards implementing them in an effective way.

Power for small devices

With technologies like Bluetooth Low Energy (BLE) coming into picture, there have been many new advancements in extremely power-saving ICs, sensors, and radio technologies. For designing energy harvesters or alternative storage devices for such low-power systems, understanding power consumption is critical.

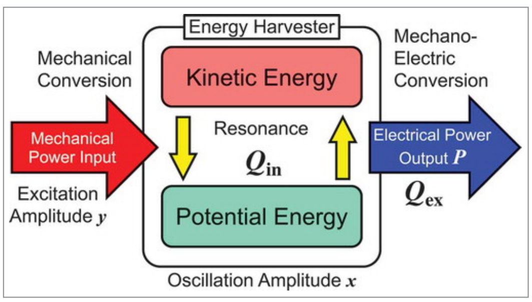

Fig. 1: Mechanism of vibrational energy harvester

The amount of energy a device requires depends on the system performance and the time taken by it to capture and transmit data. Bluetooth Low Energy and similar protocols, such as ZigBee Green Power, are optimised for short frame duration and low transmission power. Such devices are designed to transmit a complete data frame in a few milliseconds.

Sleep mode and deep sleep mode play an important role here. The longer the device stays awake, because of high energy consumption the complexity to make an energy harvester increases for such devices. If a sensor based subsystem requires 3mA at 1V, then the energy budget would be 30μJ per process. The important thing here is that the harvester must be able to supply reliably throughout its entire activity cycle. If this activity interval is between one and ten seconds, the harvester source must operate effectively during that interval.

Majority of IoT devices rely on cellular networks. With vast amounts of data across these networks, the rate of processing is slow and high latency is created. Latency here refers to the time delay from when data is sent from a connected device till it returns to the same device. Low latency is desired for IoT networks, but it comes at a price of high power usage. More energy is needed for faster data transfer. Hence, designers must balance speed and energy consumption.

Energy harvesting techniques

Energy harvesting refers to harnessing small amounts of ambient energy present in environment to power low-power electronic devices. One of the earliest applications was of crystal based radio receiver where power of the received radio signals is used